

A NEW EDGE-ADAPTIVE ZOOMING ALGORITHM FOR DIGITAL IMAGES

S. BATTIATO, G. GALLO, F. STANCO

Dipartimento di Matematica ed Informatica - Università di Catania
Viale A. Doria 6 - 95125 Catania, Italy
{battiato, gallo, fstanco}@dmi.unict.it

ABSTRACT

In this paper we address the problem of producing an enlarged picture from a given digital image (zooming). We propose a method that tries to take into account information about discontinuities or sharp luminance variations while doubling the input picture. This is realized by a multiple scan of the zoomed image and could hence be implemented with limited computational resources. The algorithm works on monochromatic images, but it is easily adapted to zoom RGB color pictures.

Our experiments show that the proposed method beats in quality pixel replication and bilinear interpolation. Moreover our algorithm is competitive both for quality and efficiency with bicubic interpolation.

KEYWORDS

Image Processing, Zooming, Interpolation, Non linear DSP.

INTRODUCTION

In this paper we address the problem of producing an enlarged picture from a given digital image (zooming). This problem arise frequently whenever an user wish to zoom in to get a better view of a given picture. There are several issues to take into account about zooming: unavoidable smoothing effects, reconstruction of high frequencies details without the introduction of artifacts and computational efficiency both in time and in memory requirements.

Several good zooming techniques are nowadays well known [3, 4, 5, 6, 7, 10, 13, 14]. A generic zooming algorithm takes as an input a RGB picture and provides as output a picture of double size preserving as much as possible the information content of the original image. For a large class of zooming techniques this is achieved by mean of some kind of interpolation: replication, bilinear and bicubic are the most popular choices and they are routinely implemented in commercial digital image processing software.

Unfortunately, these methods, while preserving the low frequencies content of the source image, are not equally able to enhance high frequencies in order to

provide a picture whose visual sharpness matches the quality of the original image. The method proposed in this paper, similarly to other published algorithms [1, 2, 14], tries to take into account information about discontinuities or sharp luminance variations while doubling the input picture. This is realized by a non linear multiple scan of the original image and could hence be implemented with limited computational resources.

The basic idea of the new algorithm is to perform a gradient-controlled, weighted interpolation. The method, however, does not require a preliminary gradient computation because the relevant information is collected during the zooming process. Although not linear, our techniques is simpler and hence much faster than fractal based zooming algorithms [10, 12].

Our experiments show that the proposed method beats in quality pixel replication and bilinear interpolation. Moreover our algorithm is competitive both for quality and efficiency with bicubic interpolation.

The rest of the paper is organized as follows. Section 1 provides a detailed description of the basic algorithm together with some considerations about its complexity. Section 2 describes how to adapt the proposed algorithm to color images. Section 3 reports the results obtained together with a discussion. A conclusion Section ends the paper.

1. THE BASIC ALGORITHM

In this Section we give a detailed description of the proposed algorithm. The procedure is summarized in pseudo-code in a following box.

First we describe the algorithm in the case of gray scale picture. A generalization to color images is described in a following Section.

The algorithm works in four successive stages.

The first stage is the simplest one and requires to expand the source $n \times n$ pixels image onto a regular grid of size $(2n - 1) \times (2n - 1)$ (see Fig. 2). More precisely if $S(i,j)$ denotes the pixel in the i -th row and j -th column of the source image, and $Z(l,k)$ denotes the pixel in the l -th row and k -th column in the zoomed picture the

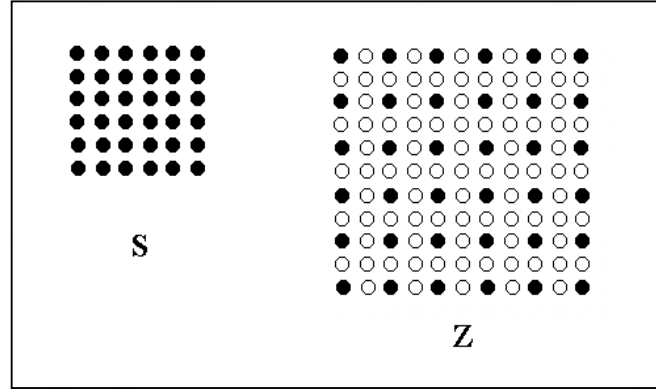


Fig. 1: The picture illustrate the first stage of zooming (expansion).

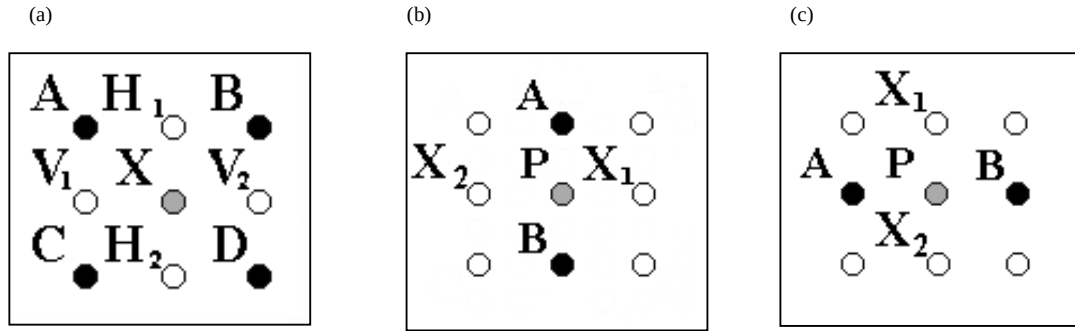


Fig. 2: (a) shows the layout of the pixel to reconstruct; (b) and (c) report the layout and the notation referred in the description of the algorithm.

expansion is described as a mapping $E:S \rightarrow Z$ according to the equation:

$$E(S(i,j))=Z(2i-1,2j-1)$$

The mapping E leaves undefined the value of all the pixels in Z with at least one even coordinate (white dots in Fig. 1).

In the second stage the algorithm scans line by line the pixels in Z whose coordinates are both even. These are the pixels denoted by a gray dot labeled with an X in Fig. 2a. For reference, in the following description we will use the capital letters A, B, C and D as in Fig. 2a, to denote the pixels surrounding the pixel X and that have already been assigned a value in the previous stage (black dots). With letters a, b, c and d we denote the luminance value of pixels A, B, C, D respectively.

For every X pixel one of the following mutual exclusive conditions is tested and a consequential action is taken:

- *(uniformity):* $\text{range}(a,b,c,d) < T_1$, where T_1 denotes a suitable threshold value; in this case the value $(a + b + c + d)/4$ is assigned to pixel X .
- *(edge in the SW-NE direction):* $|a - d| > T_2$ and $|a - d| \gg |b - c|$, where T_2 denotes a suitable threshold value; in this case the value $(b + c)/2$ is assigned to pixel X .
- *(edge in the NW-SE direction):* $|b - c| > T_2$ and $|b - c| \gg |a - d|$, where T_2 denotes a suitable threshold value; in this case the value $(a + b)/2$ is assigned to pixel X .
- *(edge in the NS direction):* $|a - d| > T_1$ and $|b - c| > T_1$ and $(a - d) * (b - c) > 0$; in this case assign the value $(a + b)/2$ to pixel H_1 and $(c + d)/2$ to pixel H_2 and leave the value at pixel X undefined.
- *(edge in the EW direction):* $|a - d| > T_1$ and $|b - c| > T_1$ and $(a - d) * (b - c) < 0$; in this case assign the value $(a + c)/2$ to pixel V_1 and $(b + d)/2$ to pixel V_2 and leave the value at pixel X undefined.

Notice that at the end of this stage there may be several X pixels, as well as several H_1 , H_2 , V_1 and V_2 pixels, left with an undefined value. Successive scan of the zoomed picture will take care of these “holes”.

In the third stage the algorithm scans line by line image Z looking for pixels left undefined in the previous stage and with at least one odd coordinate. These are the pixels denoted by the gray dot with label P in Fig. 2b and Fig. 2c. The two cases illustrated in the two sub-pictures will be treated similarly in the following discussion. For reference we will use the letters A and B as in Fig. 2b (or Fig. 2c), to denote the pixels above and below (respectively left and right) of pixel P and that have already been assigned a value in the expansion stage (black dots). With letters a and b we denote the luminance value of pixels A and B.

Observe that the values of pixels X_1 and X_2 may, or may not, have been assigned in the previous stage. If they have been assigned they will be referred respectively with x_1 and x_2 . Accordingly the algorithm considers the following two cases and take consequential actions:

- (X_1 or X_2 have not been assigned a value) In this case the algorithm checks if $|a-b| < T_1$, with the same threshold T_1 introduced in the previous stage. If it is so, the value of the pixel P is set equal to $(a + b) / 2$, otherwise the value of pixel P is left undefined.
- (X_1 and X_2 have both been assigned a value) In this case the algorithm checks for the presence of a vertical or an horizontal edge. The following sub-cases arise:
 - (presence of an edge in direction X_1X_2) $|a - b| > T_2$ and $|a - b| \gg |x_1 - x_2|$. In this case the value $(x_1 + x_2)/2$ is assigned to pixel P.
 - (presence of an edge in direction AB) $|x_1 - x_2| > T_2$ and $|x_1 - x_2| \gg |a - b|$. In this case the value $(a + b)/2$ is assigned to pixel P.
 - (none of above) no action is taken and the value of P is left undefined.

At the end of this third stage all the pixels whose spatial dependence from the neighborhood values is “simple” have been assigned a value. Using the information gathered insofar, in the next stage the remaining “holes” are eventually filled.

The fourth and final stage of the algorithm scans once more picture Z looking for pixels with undefined value and fix the “hole” using a suitably weighted mean

value as described below. Preliminarily the gray value scale (typically ranging from 0 to 255) is subdivided into a rougher scale of $m = 256/q$ “bins” B_i where q is a suitable, user selected integer value. The i-th bin includes the gray values ranging from $q(i-1)$ up to $qi-1$.

The algorithm first looks for every pixel X, with both even coordinates and with an undefined value. In this case the pixels A, B, C, D surrounding X as in Fig. 2a fall within at least one and at most four bin B_i . The representative values of these bins (typically the median) are averaged to produce a value for X. Observe that in this way the more frequent values are weighted less: this “trick” guarantees, as shown in the experiments, a better detail preservation in the zoomed picture. It is also important to note that the values considered to assign a value to X are coming from the original values in the source and hence are not affected by the smoothing operations done in the three previous stages.

Eventually the algorithm looks for every pixel P, with at least one odd coordinate whose value is still left undefined. In this case the same rebinning procedure described for X pixels is performed starting from the values A, B, X_1 , X_2 as in Fig. 2b, 2c. Observe that at this step, all four of these values have been already set.

As for the complexity of the proposed approach it is simple to observe that it requires a $O(K L)$ steps to zoom out of a factor 2 a digital picture of size K time L pixels. As for space requirement the algorithm requires only the storage space for the zoomed picture.

2. ZOOMING COLOR PICTURES

The basic algorithm described above for gray scale pictures can be easily generalized to the case of RGB colored digital images. To do so we take advantage of the higher sensitivity of human visual system to luminance variations respect to the sensitivity to chrominance values. It makes hence sense to allocate larger computational resources to zoom luminance values, while chrominance values may be processed with a simpler and more economical approach.

Accordingly we propose to operate as follows:

- Translate the original RGB picture I into the YUV color model.
- Zoom the luminance values Y according with the basic algorithm described in section one.
- Zoom the U and V values using a simpler pixel replication algorithm.
- Back translate the zoomed picture into a RGB image.

The results obtained with this basic approach are qualitatively comparable with the results obtained using bicubic interpolation over the three color channels.

```

Z= enlarge (E);

For all pixel X in Z do
  Read ( A, B, C, D)
  If A, B, C, D are uniform then
    X=(A + B + C + D)/4;
  else if there is edge SW-NE then
    X=(B + C)/2;
  else if there is edge NW-SE then
    X=(A + B)/2;
  else if there is edge NS
    H1=(A+B)/2, H2=(C+D)/2;
  else if there is edge EW
    V1=(A+C)/2, V2=(B+D)/2;
  end if
end for

For all pixel P in Z do
  Read (A, B);
  if X1, X2 are assigned then
    if A, B are uniform then
      P=(A + B)/2;
    end if;
  else if there is edge X1X2 then
    P=(X1 + X2)/2;
  else if there is edge AB then
    P=(A + B)/2;
  end if
end for

rebinning(Z);
end.

```

Box1- A pseudo code description of the algorithm.

3. EXPERIMENTAL RESULTS

The validation of a zooming algorithm requires the: assessment of the visual quality of the zoomed pictures. Unfortunately this qualitative judgement involves qualitative evaluation of many zoomed pictures from a large pool of human observers and it is hard to be done in a subjective and precise way. For this reason several alternative quantitative measurements related to picture quality have been proposed and widely used in the literature. To validate our algorithm we have chosen the approach proposed in [8, 9, 11]. In those papers the cross-correlation between the original picture and the reconstructed picture is one of the main tools used to assess the quality of reconstruction.

In our experimental contest we have first collected a test pool of 30 gray scale pictures. For each image I in

this set we have first performed the following operations:

- reduction by decimation: a new picture I_d of half the size of I is obtained taking only the pixels with both even coordinates of the original picture;
- reduction by averaging: a new picture I_a of half the size of I is obtained taking for each pixel the average value in a square of two by two pixels in the original image.

The reason for obtaining two versions of a picture as a basis for zooming, is that the size reduction technique adopted may have influences on the quality of the zoomed picture.

Starting from I_d and I_a we have obtained the zoomed images listed as follows:

- I_{dr}, the picture I_d zoomed by a factor 2 using a simple pixel replication technique;
- I_{ar}, the picture I_a zoomed by a factor 2 using a simple pixel replication technique;
- I_{db}, the picture I_d zoomed by a factor 2 using bicubic interpolation;
- I_{ab}, the picture I_a zoomed by a factor 2 using bicubic interpolation;
- I_{do}, the picture I_d zoomed by a factor 2 using our algorithm;
- I_{ao}, the picture I_a zoomed by a factor 2 using our algorithm.

We have chosen simple replication and bicubic interpolation as the two comparing stones to assess the quality of our technique. It is generally accepted that replication provides the worst quality-wise zooming algorithm while bicubic interpolation is considered one of the best options available. It should also be observed that our technique requires the choice of two threshold parameters, that could be, in principle, user-selected. We have performed our tests with a fixed default setting of the thresholds (T₁=4 T₂=3). These values have been selected because they provided, in preliminary experiments the best zooming quality. Preliminary experiments, moreover, have shown that small variations in these parameters do not produces large quality changes in the zoomed picture.

Let the cross-correlation coefficient C between two pictures A, B, be:

$$C = | (\sum_{k,l} A(k,l) B(k,l) - K L a b) / ((\sum_{k,l} A^2(k,l) - K L a^2) (\sum_{k,l} B^2(k,l) - K L b^2))^{1/2} |$$

In the above formula *a* and *b* denote respectively the average value of picture A and B; K and L denote respectively width and length, in pixels, of images A and B. Notice that cross-correlation coefficients is

between 0 and 1. The more the coefficient approaches 1, the better the reconstruction quality.

For every pair (I, I_{dr}) , (I, I_{db}) , (I, I_{do}) , (I, I_{ar}) , (I, I_{ab}) , (I, I_{ao}) we have computed the relative cross-correlation coefficients C_{dr} , C_{db} , C_{do} , C_{ar} , C_{ab} , C_{ao} .

The coefficients have been averaged over the test pool to obtain the average coefficients: C_{dr} , C_{db} , C_{do} , C_{ar} , C_{ab} , C_{ao} . The values obtained are reported in Table 1.

C_{do}	C_{db}	C_{dr}	C_{ao}	C_{ab}	C_{ar}
0.97988	0.95419	0.92178	0.97219	0.97386	0.97454

Table 1

The table promptly shows that the cross-correlation coefficients obtained with the proposed technique are better or very close to the analogous coefficients obtained using bicubic interpolation when the zooming is done from a decimated picture. This proves the better or equal ability of our algorithm to fill in with proper details the missing information in the larger picture. Setting to 0 the score for the pixel replication algorithm and to 1 the score for bicubic interpolation, the proposed technique scores a value of 1.792841.

The results in the case of a smaller picture obtained by averaging are essentially inconclusive because in this case the cross-correlation coefficient is not a good indicator of reconstruction quality. The results, indeed, seem to prize the pixel replication technique above the other two, even if it is very well known that this technique does not produce perceptually good images. This apparently contradictory result only comes from the statistical properties of cross-correlation coefficient. This statistical indicator turns out closer to one whenever a square of two time two pixels is substituted by a simple replication of their average. The behavior in the case of zoomed picture obtained from averaged smaller picture should hence be assessed using a different technique. To this aim we computed the percentage of zoomed pixels whose difference, in absolute value between the computed and the original intensity is greater than a given threshold.

Fig. 3a shows the average percentage of errors observed over the test pool as different tolerance threshold are considered in the case when the sub-sampled image is obtained by decimation. The proposed technique clearly outperforms the other two. Fig. 3b shows the average percentage of errors observed over the test pool as different tolerance threshold are considered in the case when the sub-sampled image is

obtained by local averaging. In this case as well the proposed technique gives a better performance.

4. CONCLUSIONS

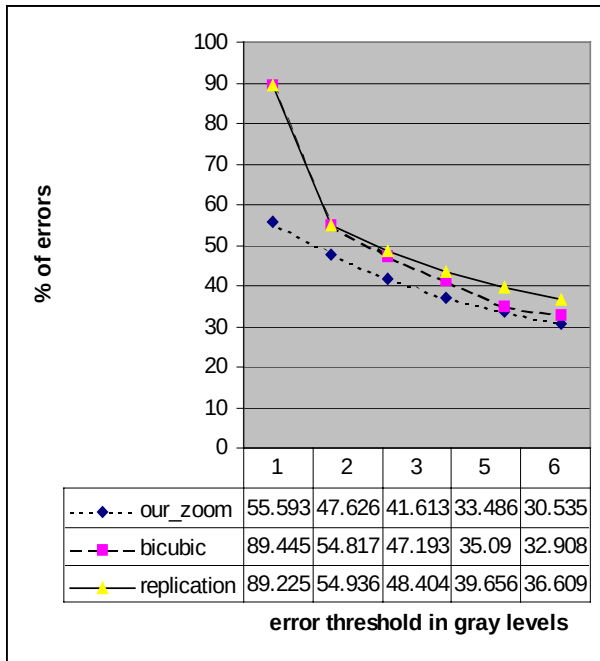
In this paper we have proposed a new technique for zooming a digital picture, both in gray scale and in RGB colors. The proposed technique has been compared according to different performance indicators to bicubic interpolation and pixel replication algorithm. The experimental results show that the proposed method, while of not greater complexity than bicubic interpolation provides qualitatively better results.

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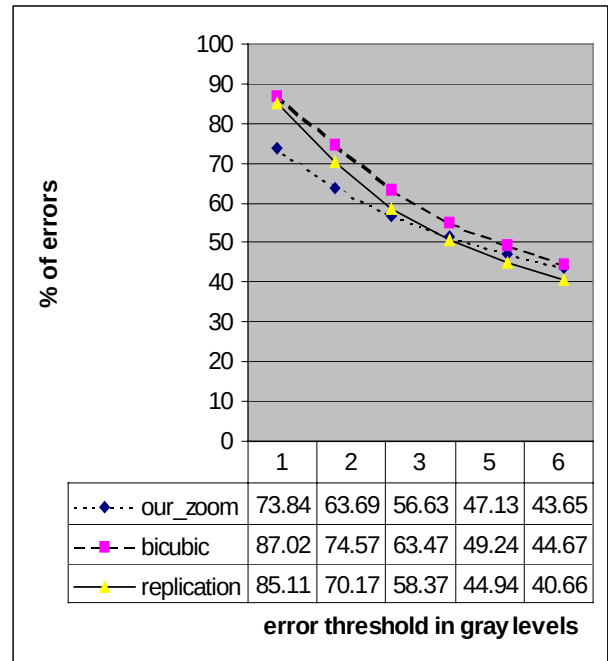
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(a)



(b)

Fig. 3: The average percentage of errors observed over the test pool for different tolerance thresholds when the sub-sampled image is obtained: (a) by local averaging; (b) by decimation

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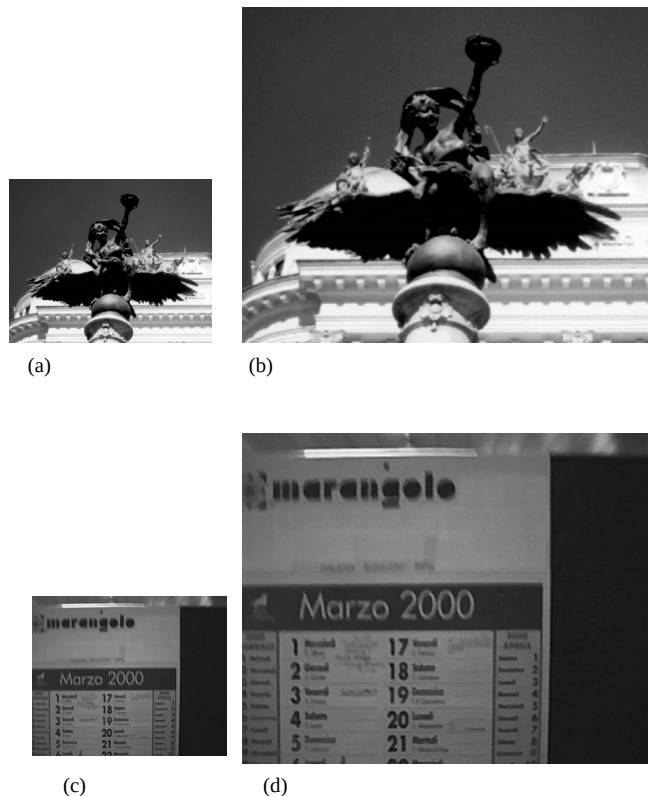


Fig. 4: Two examples of zoomed pictures obtained with the proposed algorithm. (a) and (b) relative to a natural scene, (c) and (d) relative to a printed text.